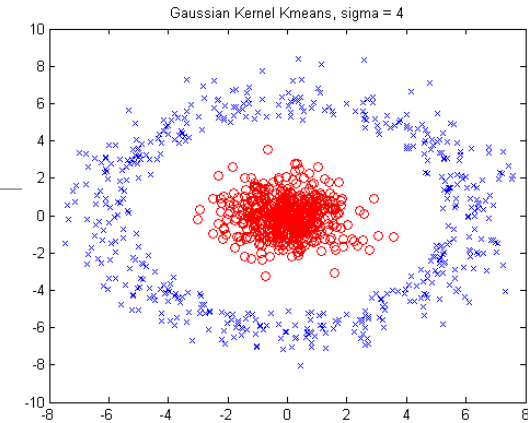


Kernel K-Means Clustering

- ❑ Kernel K-Means can be used to detect non-convex clusters
 - ❑ K-Means can only detect clusters that are linearly separable
- ❑ Idea: Project data onto the high-dimensional kernel space, and then perform K-Means clustering
 - ❑ Map data points in the input space onto a high-dimensional feature space using the kernel function
 - ❑ Perform K-Means on the mapped feature space
- ❑ Computational complexity is higher than K-Means
 - ❑ Need to compute and store $n \times n$ kernel matrix generated from the kernel function on the original data
- ❑ The widely studied spectral clustering can be considered as a variant of Kernel K-Means clustering



Kernel Functions and Kernel K-Means Clustering

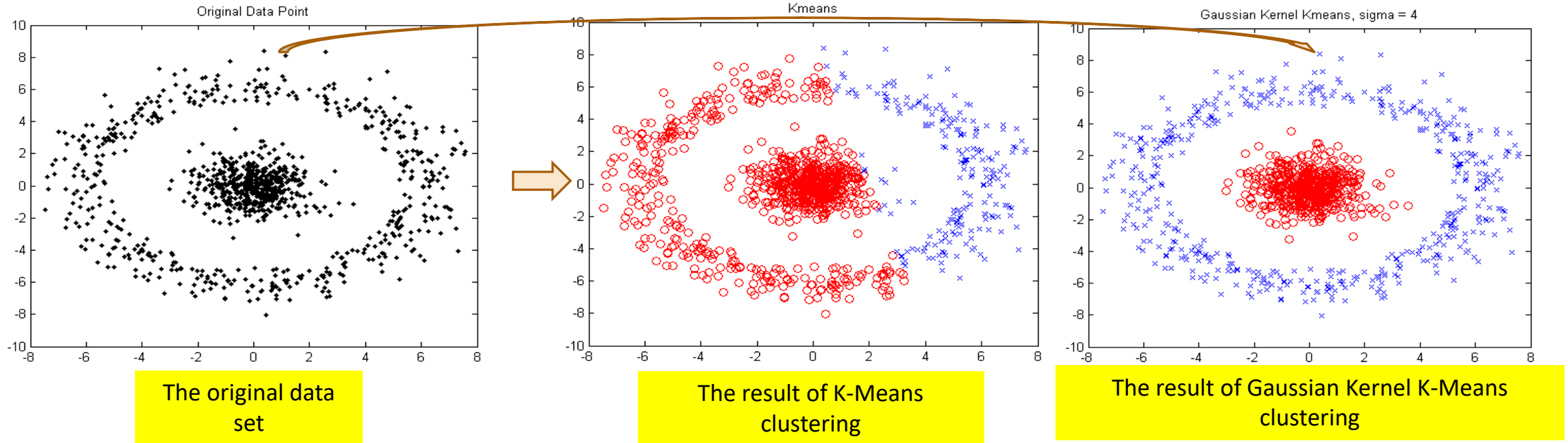
- Typical kernel functions:
 - Polynomial kernel of degree h : $K(\mathbf{X}_i, \mathbf{X}_j) = (\mathbf{X}_i \mathbf{X}_j + 1)^h$
 - Gaussian radial basis function (RBF) kernel: $K(\mathbf{X}_i, \mathbf{X}_j) = e^{-\frac{|\mathbf{X}_i - \mathbf{X}_j|^2}{2\sigma^2}}$
 - Sigmoid kernel: $K(\mathbf{X}_i, \mathbf{X}_j) = \tanh(\kappa \mathbf{X}_i \cdot \mathbf{X}_j - \delta)$
- The formula for kernel matrix K for any two points $\mathbf{X}_i, \mathbf{X}_j \in C_k$ is $K(\mathbf{X}_i, \mathbf{X}_j) = \phi(\mathbf{X}_i) \cdot \phi(\mathbf{X}_j)$
- The SSE criterion of kernel K-means: $SSE(C) = \sum_{k=1}^K \sum_{x_i \in C_k} |\phi(x_i) - c_k|^2$
 - The formula for the cluster centroid: $c_k = \frac{\sum_{x_i \in C_k} \phi(x_i)}{|C_k|}$
- Clustering can be performed without the actual individual projections $\phi(x_i)$ and $\phi(x_j)$ for the data points $x_i, x_j \in C_k$

Example: Kernel Functions and Kernel K-Means Clustering

- Gaussian radial basis function (RBF) kernel: $K(\mathbf{X}_i, \mathbf{X}_j) = e^{-\frac{|\mathbf{X}_i - \mathbf{X}_j|^2}{2\sigma^2}}$
- Suppose there are 5 original 2-dimensional points:
 $x_1(0, 0), x_2(4, 4), x_3(-4, 4), x_4(-4, -4), x_5(4, -4)$
- If we set σ to 4, we will have the following points in the kernel space
 - E.g., $\|x_1 - x_2\|^2 = (0 - 4)^2 + (0 - 4)^2 = 32$, therefore, $K(x_1, x_2) = e^{-\frac{32}{2 \cdot 4^2}} = e^{-1}$

Original Space			RBF Kernel Space ($\sigma = 4$)				
	x	y	$K(x_i, x_1)$	$K(x_i, x_2)$	$K(x_i, x_3)$	$K(x_i, x_4)$	$K(x_i, x_5)$
x_1	0	0	1	$e^{-\frac{4^2+4^2}{2 \cdot 4^2}} = e^{-1}$	e^{-1}	e^{-1}	e^{-1}
x_2	4	4	e^{-1}	1	e^{-2}	e^{-4}	e^{-2}
x_3	-4	4	e^{-1}	e^{-2}	1	e^{-2}	e^{-4}
x_4	-4	-4	e^{-1}	e^{-4}	e^{-2}	1	e^{-2}
x_5	4	-4	e^{-1}	e^{-2}	e^{-4}	e^{-2}	1

Example: Kernel K-Means Clustering



- ❑ The above data set cannot generate quality clusters by K-Means since it contains non-convex clusters
- ❑ Gaussian RBF Kernel transformation maps data to a kernel matrix K for any two points x_i, x_j : $K_{x_i x_j} = \phi(x_i) \bullet \phi(x_j)$ and Gaussian kernel: $K(x_i, x_j) = e^{-\|x_i - x_j\|^2 / 2\sigma^2}$
- ❑ K-Means clustering is conducted on the mapped data, generating quality clusters